Week 4- Report Threat Hunting Simulation

Summary

In this investigation, a multi-step cyber attack was identified. The attacker gained access to a user account, placed a malicious file in the system, and used PowerShell to perform suspicious activity.

Identified IOCs

- **Attacker IP:** 10.10.50.200
- **Malicious File:** mimikatz.exe
- **File Location:** C:\temp\
- **File Hash:** 1d3cb39d216973fe80c2557640238d6e77165a52ea9661769482fe3112962bd9
- **PowerShell Command:** powershell -nop -w hidden -enc ...

Attack Timeline

1. **Brute Force Attack:**
   The attacker used IP 10.10.50.200 to try multiple login attempts and gain access to the system.
2. **Malicious File Drop:**
   After login, the attacker copied a file named mimikatz.exe into the system. This file is commonly used to steal passwords.
3. **PowerShell Activity:**
   PowerShell was used with encoded and hidden commands, which indicates suspicious activity and possible data transfer.

Analysis

The attack followed a clear pattern:

- First, the attacker entered the system using brute force
- Then, a malicious file was placed inside the system
- Finally, PowerShell was used to perform hidden actions

This shows a complete attack flow from access to execution.

Impact

- User credentials may be stolen
- System security is compromised
- Sensitive data may be at risk

Recommendations

- Use strong passwords and enable account lockout
- Monitor PowerShell usage regularly
- Block suspicious tools like mimikatz
- Enable security monitoring tools (SIEM/EDR)
- Review logs frequently

Conclusion

This investigation confirms a real attack. The attacker successfully accessed the system, executed a malicious file, and used PowerShell for suspicious activity. Immediate action is required to secure the system.